CMPn332 – CMPN332 – CMP432

Linear transformation(image negative)	Logarithmic Transformation	Power transformation (gamma correction)
Used for enhancing white or grey detail embedded in dark regions of an image. The transformation function used in image negative is → **S**= $L-1-\boldsymbol{r}$.	Enhancing image details in darker regions while compressing lighter areas The transformation function used is $\boldsymbol{S=c*log\,(1+r)}$ • c defined to normalize the resulting image to a valid (visible) range→$c=\frac{L-1}{log(L)}$	Used to map a narrow range of dark input values into a wider range of output values or vice versa depending on γ value The transformation function used is→ $\boldsymbol{S=c*r^{\gamma}}$

-
`int main()`
`{`
`Mat src = imread("neg.jpg", 0);`
`namedWindow("src", 0);`
`imshow("src", src);`
`for (int i = 0; i < src.rows; i++)`
`for (int j = 0; j < src.cols; j++)`
`src.at<uchar>(i, j) = 255 - src.at<uchar>(i, j);`
`namedWindow("negative", 0);`
`imshow("negative", src);`
`waitKey(0);`
`}`
	`Mat src = imread("log1.jpg", 0);`
	`Mat dst(src.rows,src.cols, CV_32FC1);`
	`namedWindow("Src Image", 0);`
	`imshow("Src Image", src);`
	`waitKey(0);`
	`for (int i = 0; i < src.rows; i++)`
	`{`
	`for (int j = 0; j < src.cols; j++)`
	`{`
	`dst.at<float>(i, j) = log(src.at<uchar>(i, j) + 1);`
	`}`
	`}`
	`normalize(dst, dst, 0, 255, NORM_MINMAX);`
	`convertScaleAbs(dst, dst);`
	`namedWindow("out", 0);`
	`imshow("out", dst);`
	`waitKey(0);`
		`Mat src = imread("log1.jpg", 0);`
		`Mat dst(src.rows, src.cols, CV_32FC1);`
		`namedWindow("Src Image", 0);`
		`imshow("Src Image", src);`
		`waitKey(0);`
		`for (int i = 0; i < src.rows; i++)`
		`{`
		`for (int j = 0; j < src.cols; j++)`
		`{`
		`dst.at<float>(i, j) = powf(src.at<uchar>(i, j), 1 );`
		`}`
		`}`
		`normalize(dst, dst, 0, 255, NORM_MINMAX);`
		`convertScaleAbs(dst, dst);`
		`namedWindow("out", 0);`
		`imshow("out", dst);`
		`waitKey(0);`

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Bit Plane Slicing

The main goals of bit plane slicing is:

- Converting a gray level image to a binary image.
- Image Compression (Representing an image with fewer bits)

```
Mat src_img = imread("bit.jpg", 0);
namedWindow("src_img", 0);
imshow("src_img", src_img);
for (int i = 0; i < src_img.rows; i++)
    for (int j = 0; j < src_img.cols; j++)
    {
        if (src_img.at<uchar>(i, j) & 64)
            src_img.at<uchar>(i, j) = 255;
        else src_img.at<uchar>(i, j) = 0;
    }
namedWindow("bit_plane", 0);
imshow("bit_plane", src_img);
```

Gray level slicing

It's a way to highlight gray range of interest to a viewer in one of two ways:

- Transformation highlights range [A, B] of gray level and reduces all others to a constant level

  $S = 255\ if\ A > r < B, C\ otherwise$

- Transformation highlights range [A, B] but preserves all other levels

  $S = 255\ if\ A > r < B, \text{input}\ otherwise$

```
Mat src_img = imread("g.jpg", 0);
namedWindow("src_img", 0);
imshow("src_img", src_img);
for (int i = 0; i < src_img.rows; i++)
    for (int j = 0; j < src_img.cols; j++)
    {
        if (src_img.at<uchar>(i, j) > 130 && src_img.at<uchar>(i, j) < 200)
            src_img.at<uchar>(i, j) = 255;
        else src_img.at<uchar>(i, j) = src_img.at<uchar>(i, j);
    }
namedWindow("bit_plane", 0);
imshow("bit_plane", src_img);
waitKey(0);
}
```

Gamma correction output→ $s=c*r^{\gamma}$

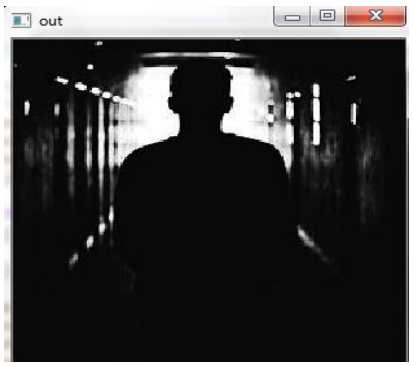
γ=1



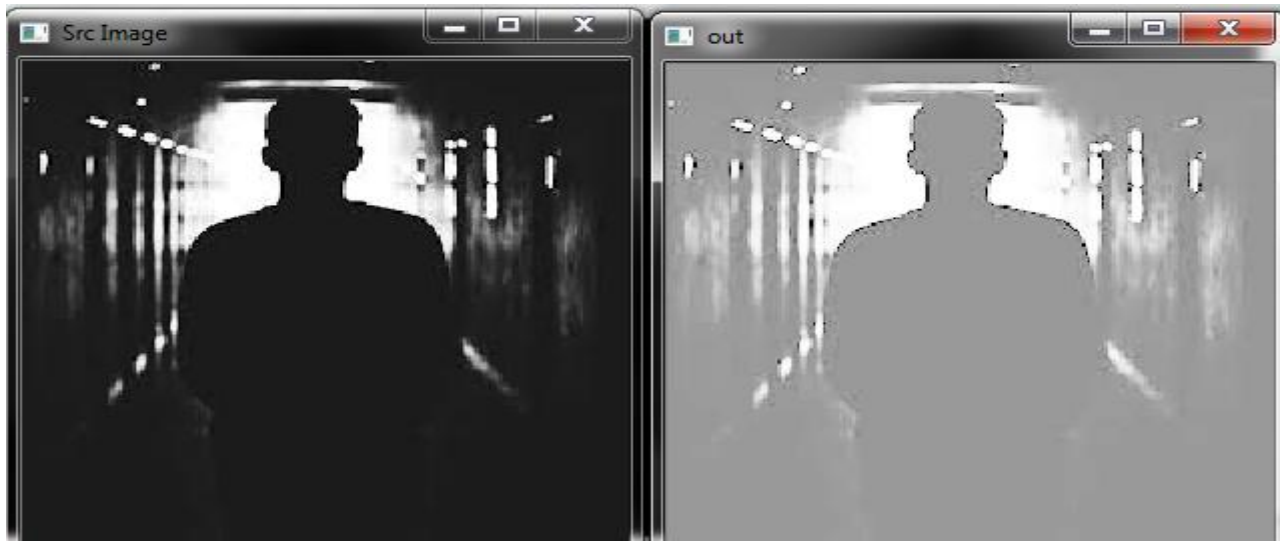
γ=0.5



γ=1.5



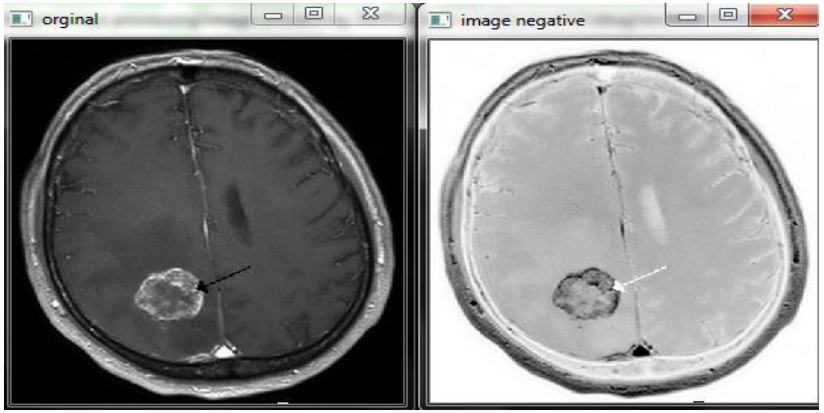
Logarithmic transformation output→$s = c * log_e(1 + r)$



Bit Plane slicing output



Image Negative output → S=intensity $_{max} - r$.



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Giving the following image F(x,y)

205	**99**	**67**
156	55	43
234	12	89
100	200	55

1. Which image enhancement technique used to enhance white or grey detail embedded in dark regions of an image, show enhancement effect at pixel (3,1).
2. Which image enhancement technique used to enhance image details in darker regions while compressing lighter areas, show enhancement effect at pixel (2,0).
3. Which image enhancement technique used to enhance image with high brightness, show enhancement effect at pixel (0,0).
4. Which image enhancement technique used to highlight gray range of interest to a viewer, show enhancement effect at pixel (3,2) for Range 20 to 80.
5. Bit plane slicing is a method of representing an image with different planes which plane represent the most significant data. Show its content for pixel (1,1)

Answer:

1. Image negative → S= (L-1) – r
   - For pixel (3,1) S= 255-200=55
2. Log transformation→ S = c * log (1 + r) where c= (L - 1) / log(L)
   - For pixel (2,0) S=$\frac{255}{\log 256} * \log(1 + 234) = 251$
3. Power law→$\boldsymbol{s = c * r^{\gamma}}$ $where\ c = \frac{255}{255^{\gamma}}$
   - For pixel (0,0) $S = \frac{255}{255^{1.5}} * 205^{1.5} = 184$
4. Gray Level slicing → $S(x,y)\begin{cases} 255 & \text{if } 20 < r(x,y) < 80 \\ C & \text{otherwise} \end{cases}$
   - For pixel (3,2) $S = 255$
5. Plane represents most significant data for 8bit image is Most significant pane 2^7 plane.
   - For pixel (1,1) = 0